

Basic Principles of Medicine I
Module: Cardiovascular, Pulmonary, Renal | Lecture CPR 01

Mediastinum

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Recommended Reading

- Drake Richard L, Vogl A. Wayne Mitchell Adam W. M. Gray's Anatomy for Students. 5th Ed.
- Chapter 3
 - Regional Anatomy
 - Mediastinum
 - Anterior mediastinum
 - Superior mediastinum
 - Posterior mediastinum
- Please be sure to review the 'In the Clinic' boxes for clinical correlations for this topic



Objectives

SOM.MK.II.BPM1.3.CPR.2.ANAT.1363

Describe the subdivisions of the mediastinum and their boundaries.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1364

List the structures found in the superior, anterior, middle and posterior mediastinum and their relationships to one another.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1365

Describe the relationships of the esophagus, trachea and thoracic duct in the superior mediastinum.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1366

Describe the typical branching pattern of the aortic arch in the superior mediastinum.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1367

Describe the subclavian artery and its branches.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1368

Explain what clinical impairments could be caused by the presence of a cervical rib.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1369

Describe the relationships of the phrenic nerves to the aortic arch and the root of the lung.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1370

Describe the relationships of the vagus nerves to the aortic arch, the root of the lung, and the esophagus.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1371

Describe the course of the right and left recurrent laryngeal nerves in the thorax.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1372

List the structures that are found at the plane of Ludwig.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1373

Describe the four normal constrictions of the esophagus.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1374

Describe the arterial supply, innervation, venous and lymphatic drainage of the thoracic and abdominal portions of the esophagus.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1375

Describe the azygos system of veins and their typical pattern of drainage.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1376

Describe the origin of the greater, lesser and least splanchnic nerves.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1377

List the normal openings in the diaphragm and their approximate vertebral level.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1378

Describe the anatomical location, mechanism, and clinical consequences of thoracic duct laceration.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1379

Describe the clinical signs, causes, and presentation of thoracic outlet syndrome.

SOM.MK.II.BPM1.3.CPR.2.ANAT.1383

List the pericardial coverings of the heart

SOM.MK.II.BPM1.3.CPR.2.ANAT.1384

Describe the extent and attachments of the fibrous pericardium (pericardial sac)

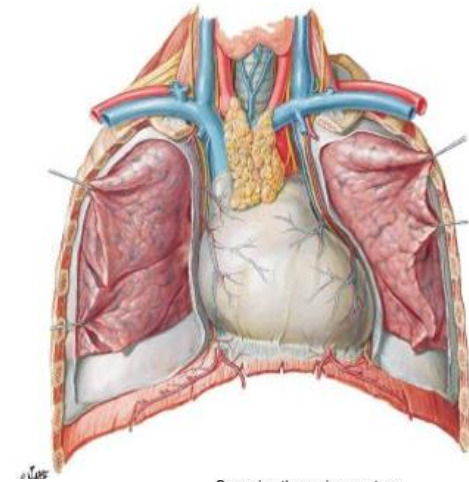
SOM.MK.II.BPM1.3.CPR.2.ANAT.1385

Describe the innervation and arterial supply of the fibrous pericardium

What is the mediastinum?

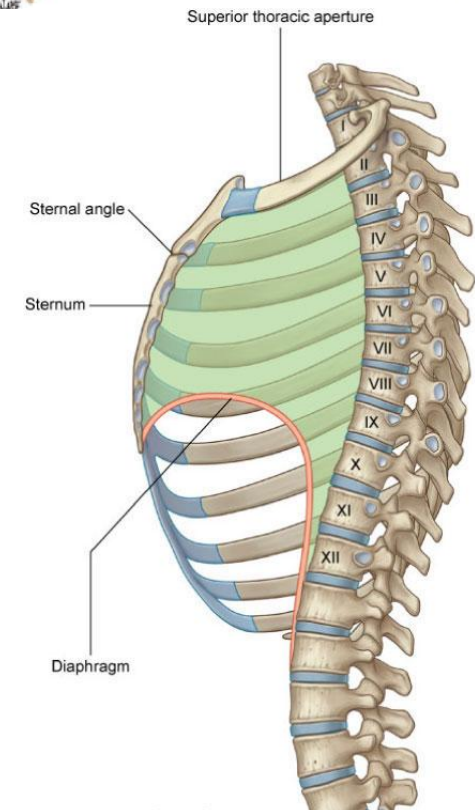
Mediastinum

- The thoracic compartment between the two pleural sacs.
- Contains the major organs of the thoracic cavity: heart, esophagus, trachea, major nerves, and major systemic blood vessels
 - excludes the lungs



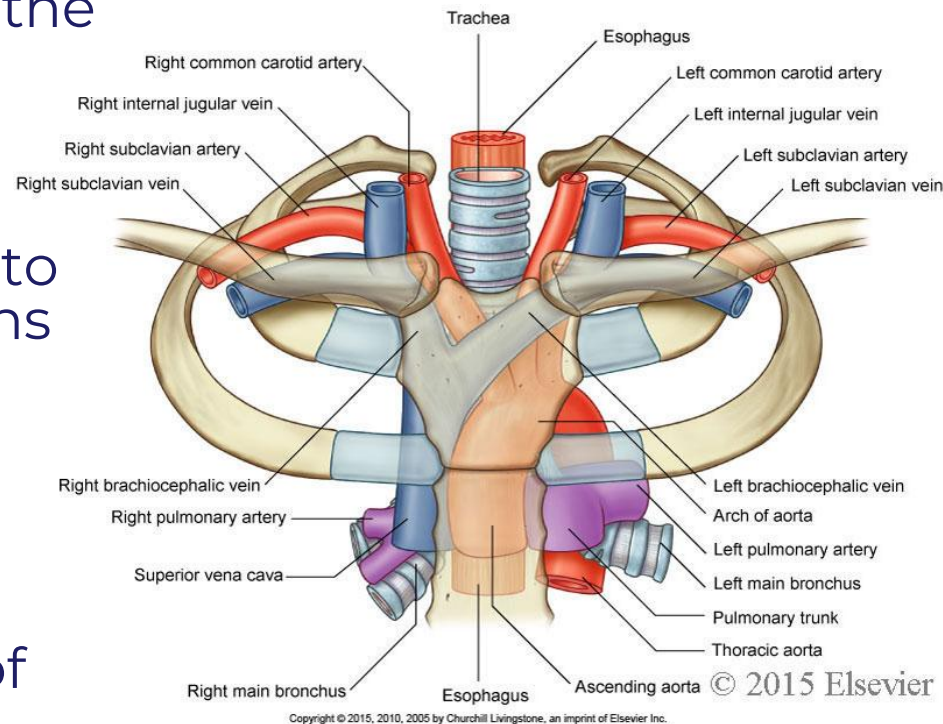
Boundaries

- *Superior*
 - Superior thoracic aperture
- *Inferior*
 - Diaphragm
- *Anterior*
 - Sternum
- *Posterior*
 - Thoracic Vertebrae

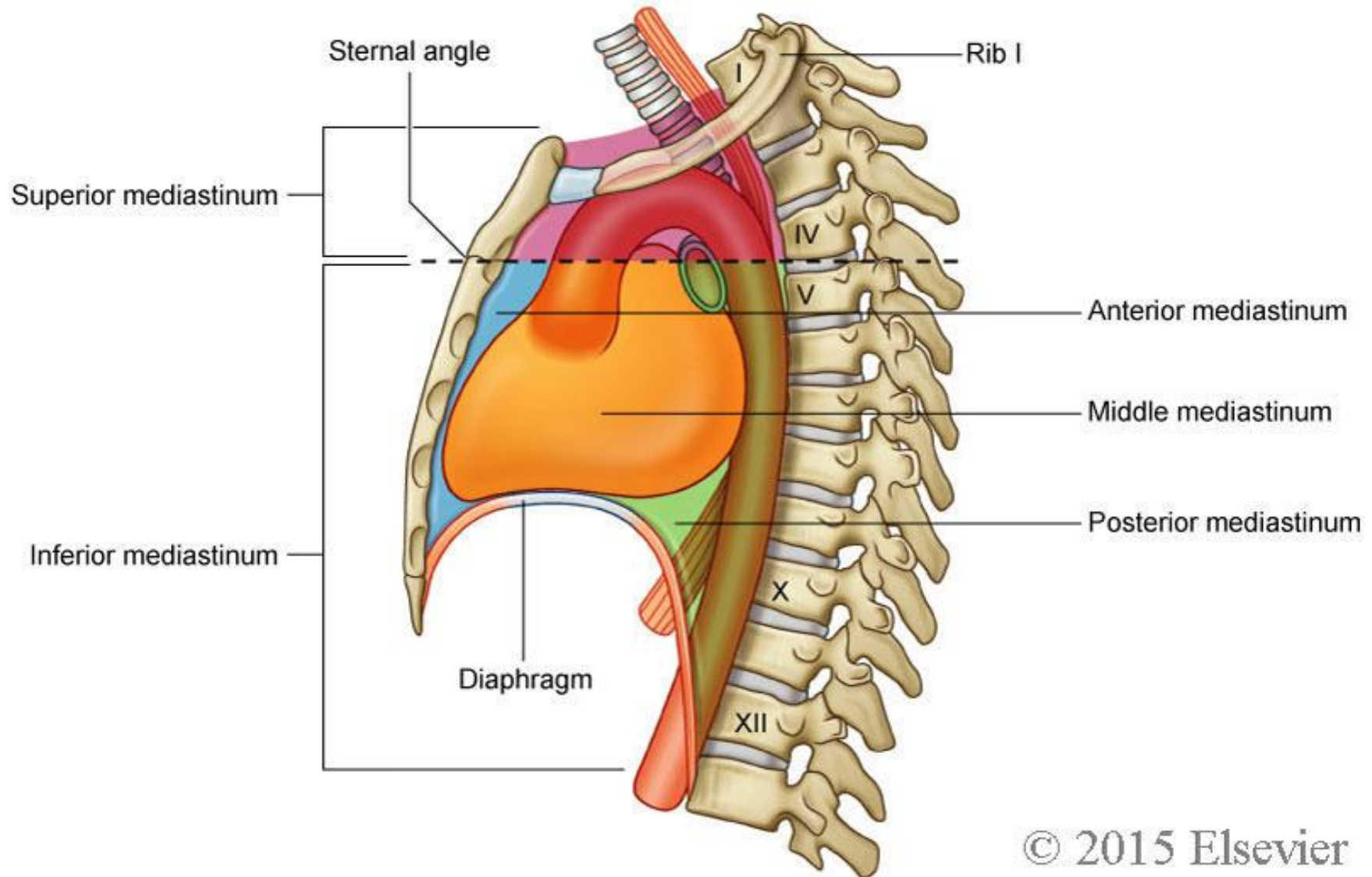


Plane of Ludwig

- i.e. transthoracic plane
- artificial horizontal plane at the level of:
 - T4/T5 posteriorly
 - second rib anteriorly
- Divides the mediastinum into superior and inferior divisions
- Structures:
 - Superior vena cava as it enters the pericardium
 - Beginning and ending of the arch of the aorta
 - Trachea bifurcation
 - Pulmonary trunk bifurcation



Divisions of the Mediastinum



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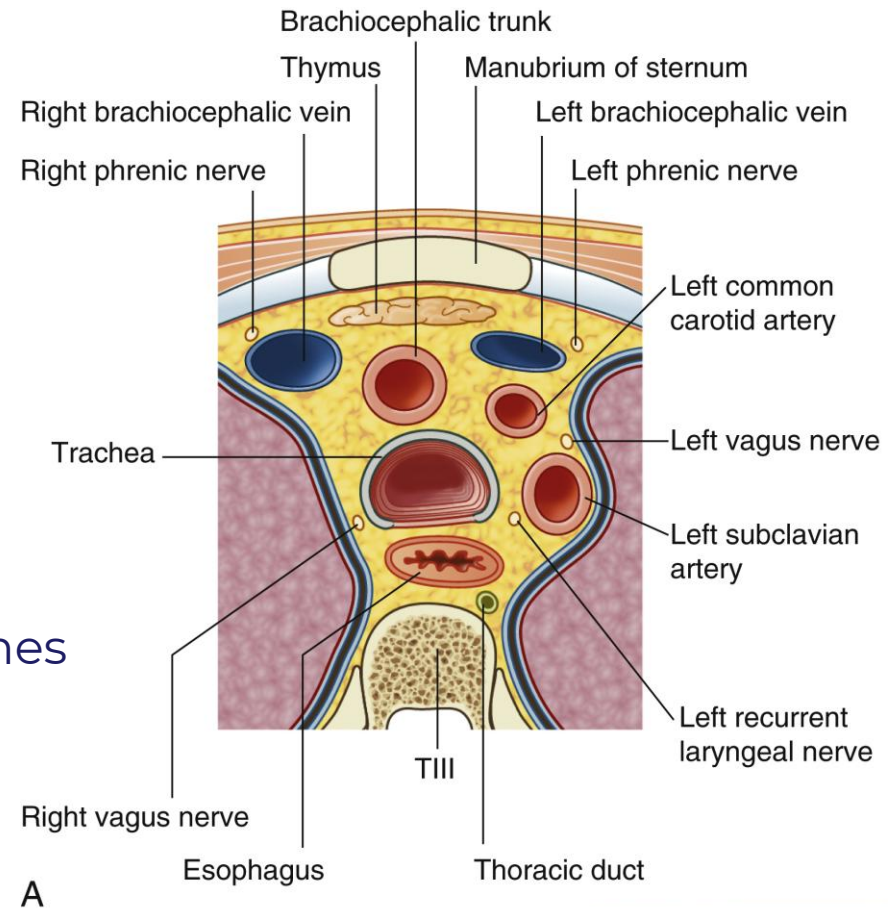
Superior Mediastinum

• Boundaries:

- Superior: 1st rib, T1, jugular notch
- Inferior: transthoracic plane
- Posterior: T1 – T4 Vertebrae
- Anterior: Manubrium

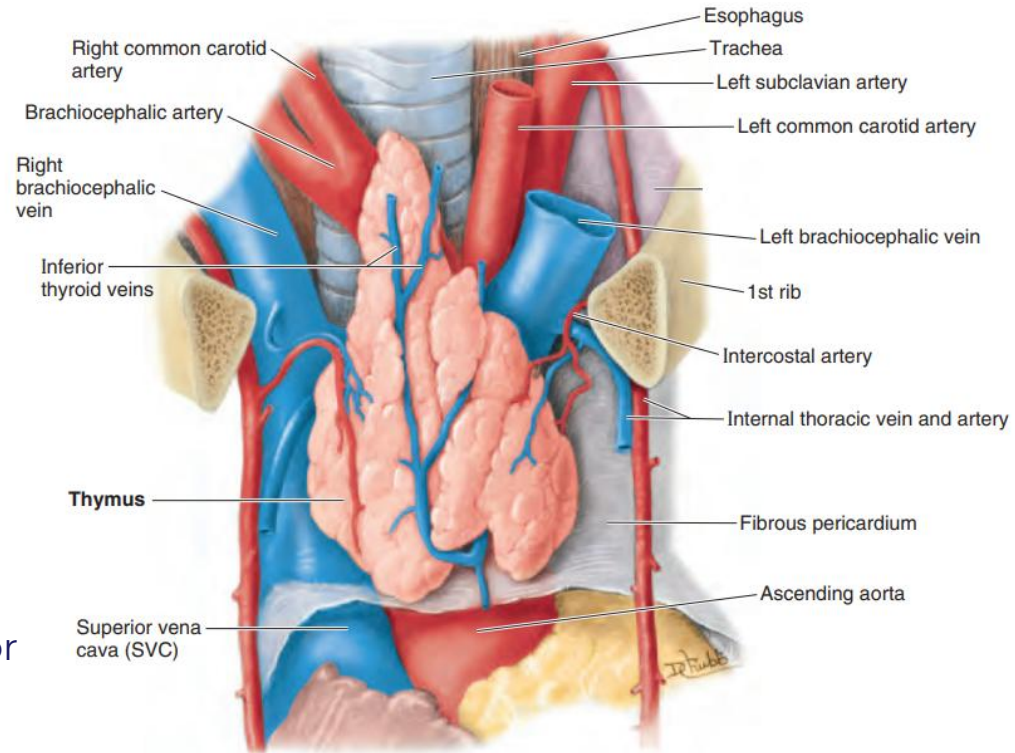
• Contents:

- Thymus***
- Trachea
- Arch of the Aorta and branches
- Brachiocephalic veins
- Superior vena cava
- Esophagus***
- Thoracic Duct***
- Nerves***



Thymus

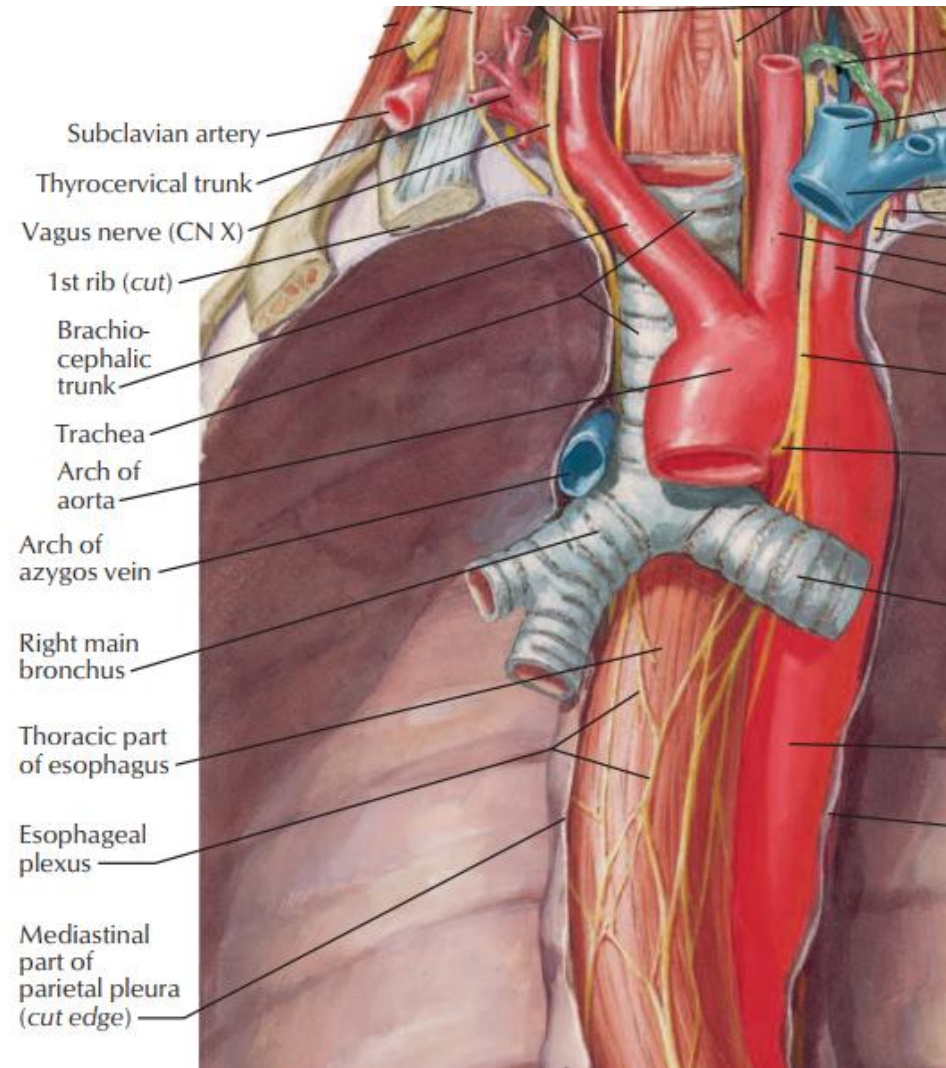
- Develops from 3rd pharyngeal pouch
 - Involutes after puberty
- Location
 - anterior part of superior mediastinum
 - Upper part extends into neck
 - Lower part extends into anterior mediastinum



- Arterial supply: internal thoracic and anterior intercostal arteries
- Venous drainage: into internal thoracic, brachiocephalic, and inferior thyroid veins
- Lymphatic drainage: into brachiocephalic, tracheobronchial, and parasternal lymph nodes

Trachea

- Continuation from the larynx at C6
- Lies anterior to the esophagus
- Terminates at the plane of Ludwig
- Bifurcates into right and left main bronchi
- Crossed over by the **arches of the aorta and the azygos vein**

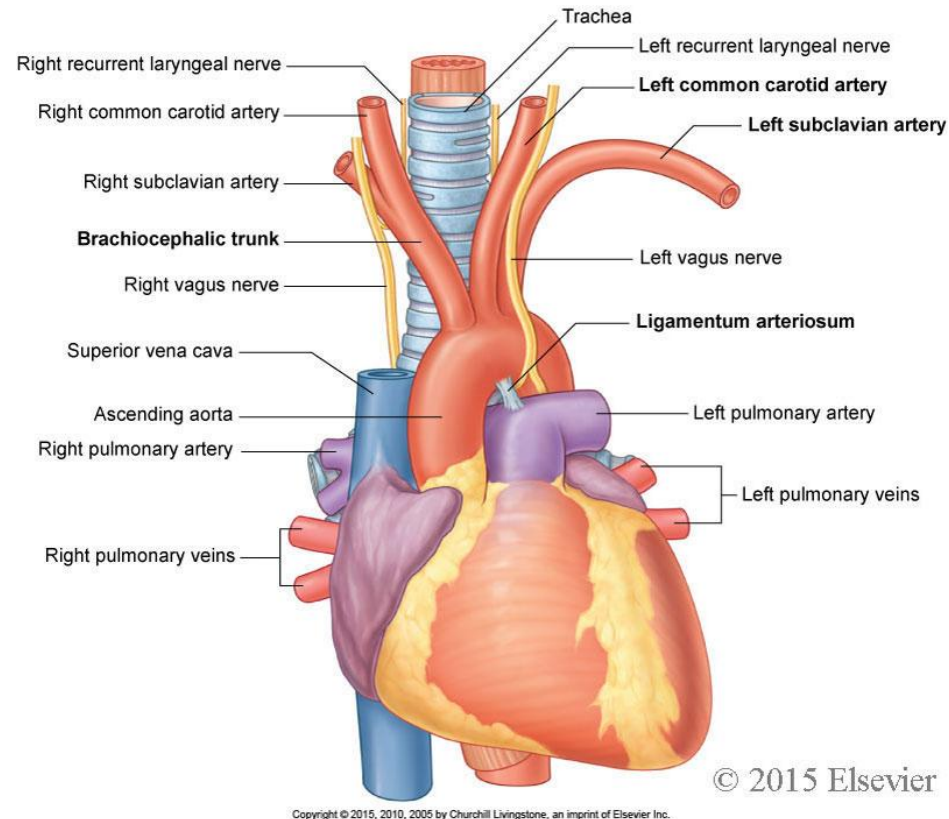


Arch of the Aorta

Branches of the Aortic Arch

- Brachiocephalic trunk
 - Right subclavian artery
 - Right common carotid artery
- Left common carotid artery
- Left subclavian artery

All three are crossed anteriorly by the left brachiocephalic vein



Brachiocephalic Veins

- Formed by the union of the subclavian and internal jugular veins
 - Located posterior to the sternoclavicular joints and the thymus
- Left brachiocephalic vein crosses the midline
 - Clinically important – must remember if performing a tracheotomy

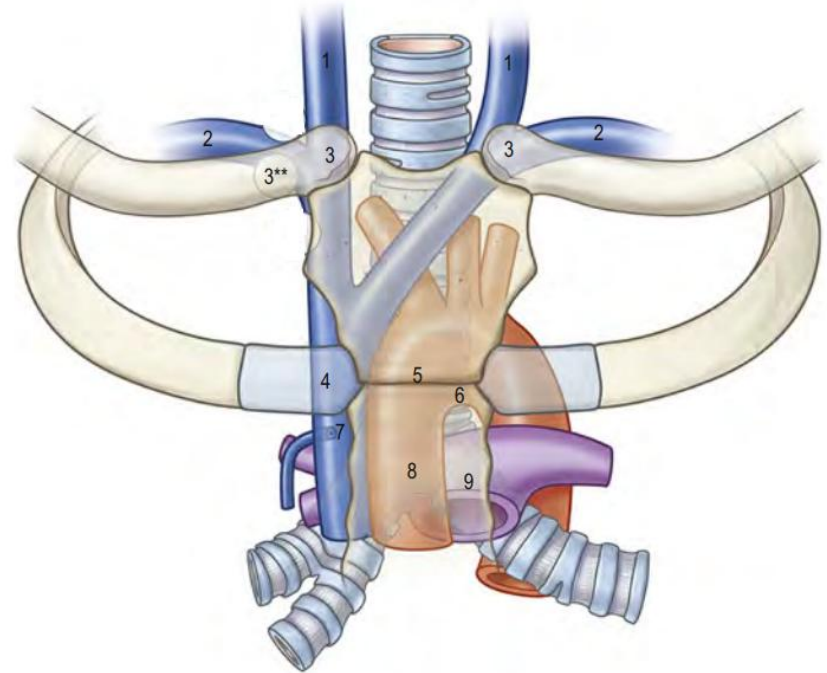
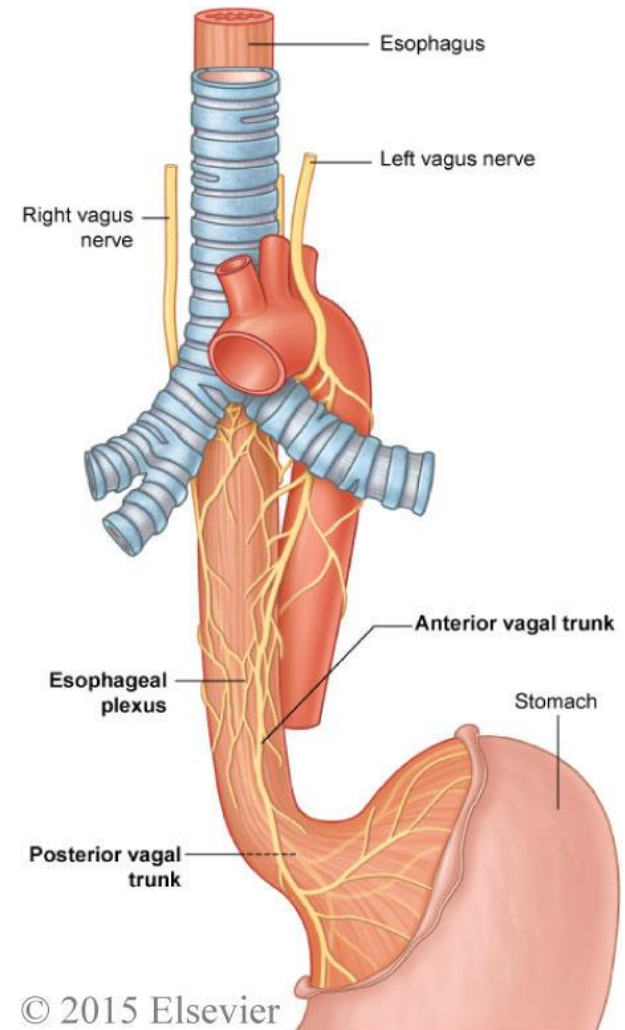


Fig. 52.7 Surface anatomy of the great vessels and tracheobronchial tree relative to bony landmarks. Key: 1, internal jugular vein; 2, subclavian vein; 3, formation of the brachiocephalic vein: commonly posterior to the sternoclavicular joint

Nerves of the mediastinum: Vagus

- Supply thoracic viscera with **parasympathetic** and **visceral afferent** innervation
- Contribute to the cardio-pulmonary plexuses
- Contribute to esophageal plexuses
 - Formed as each nerve divides into several branches that spread over the viscera
 - Just above the diaphragm these fibers converge to form **two trunks**:
 - Anterior Vagal trunk: mainly from the **left** vagus nerve.
 - Posterior Vagal trunk: mainly from the **right** vagus nerve.



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Nerves of the mediastinum: Recurrent Laryngeal

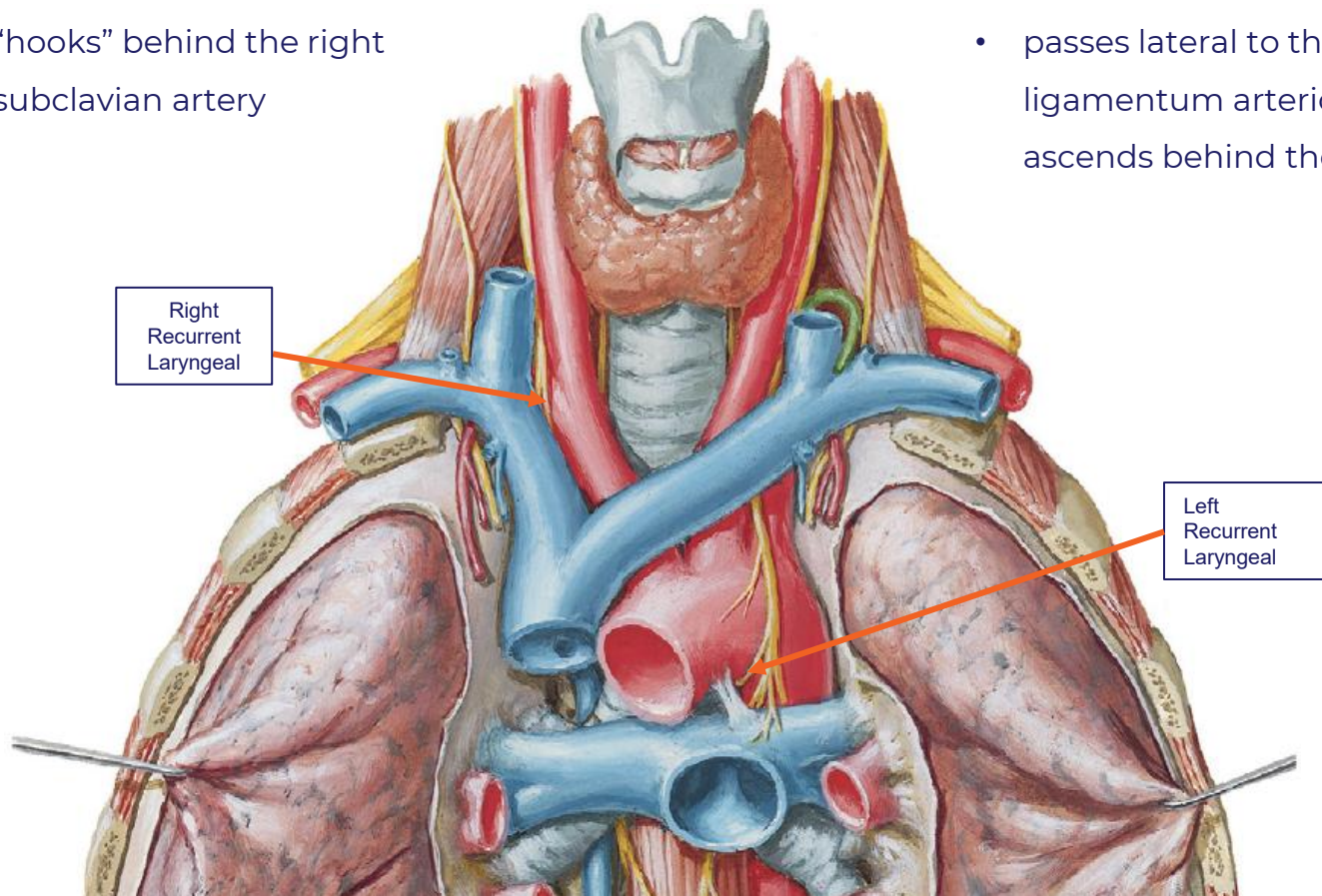
- Arise from vagus nerves on either side.
- Innervate intrinsic muscles of the ***larynx***
 - injury can present with hoarseness or loss of voice

Right recurrent laryngeal nerve:

- “hooks” behind the right subclavian artery

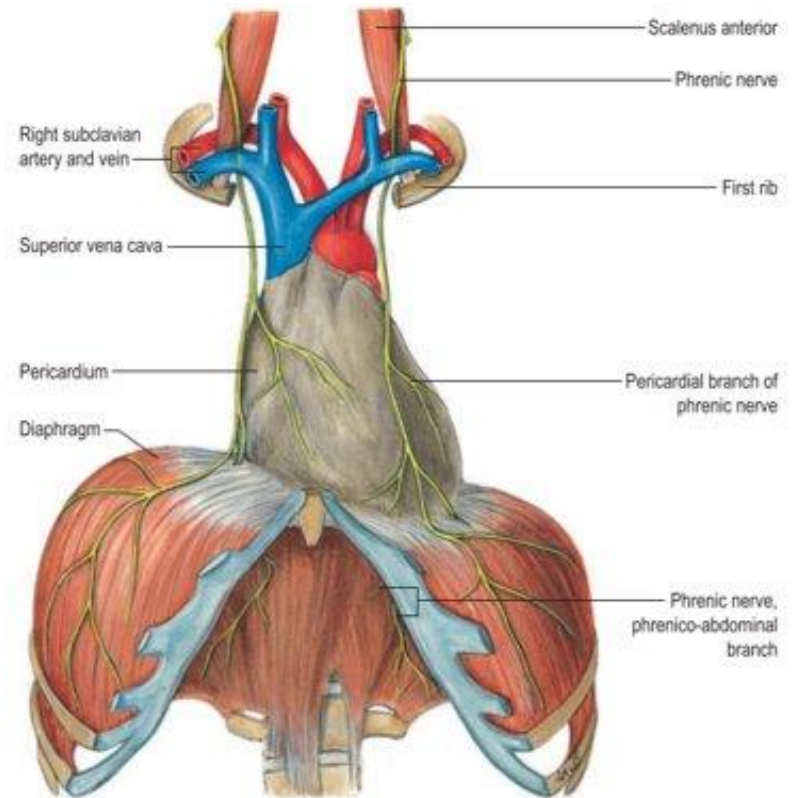
Left recurrent laryngeal nerve

- passes lateral to the ligamentum arteriosum then ascends behind the aortic arch



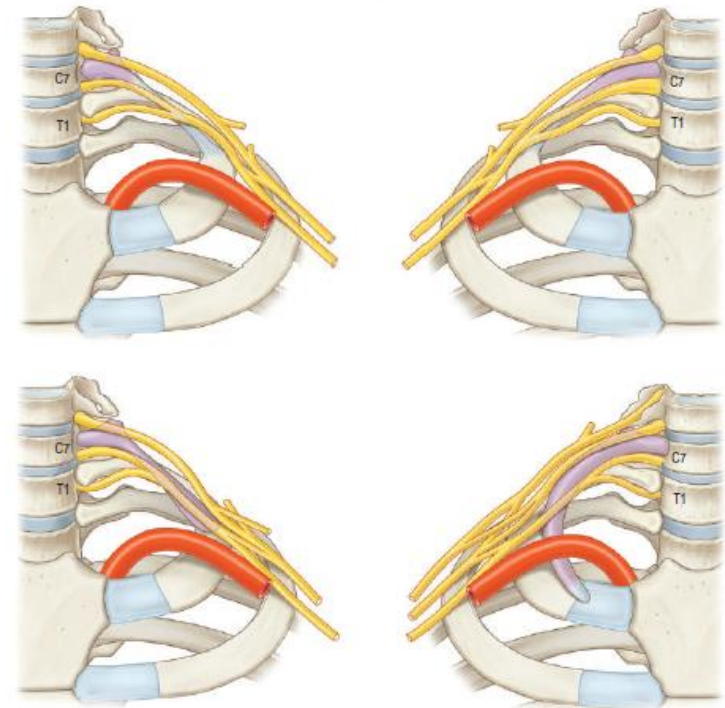
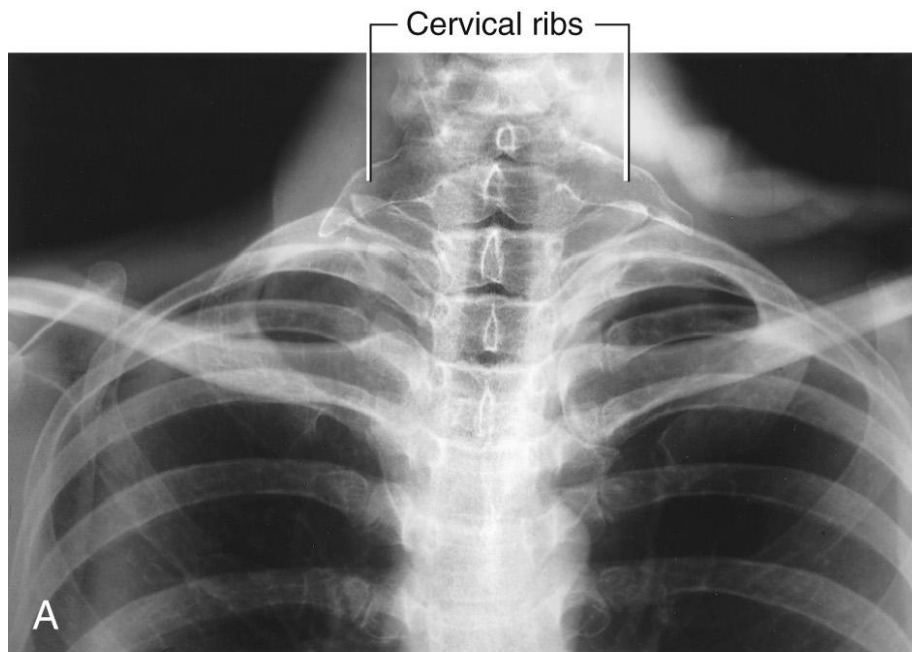
Nerves of the mediastinum: Phrenic

- **Sensory** innervation of:
 - *mediastinal pleura*
 - *fibrous pericardium*
 - *parietal layer of serous pericardium*
- **Sensory and motor** innervation to the diaphragm and associated membranes



Cervical Ribs

- Articulate posteriorly with vertebrae **C7** and anteriorly with the superior border of the anterior aspect of **rib I**
- Structures that normally pass over the rib are elevated
 - Cervical ribs may produce **thoracic outlet syndrome**



Neck radiograph with bilateral cervical ribs (may resemble horn-like structures)

Thoracic Outlet Syndrome

- **Arterial compression**
 - may coincide with atheroma and eventually distal embolization
- **Subclavian vein compression**
 - may induce axillary vein thrombosis
- **Nerve tension (C8 to T1)**
 - may lead to sensory disturbances and muscle wasting

Note: Thoracic outlet syndrome can be also caused by conditions such as muscular abnormalities, trauma, or tumors in the region.



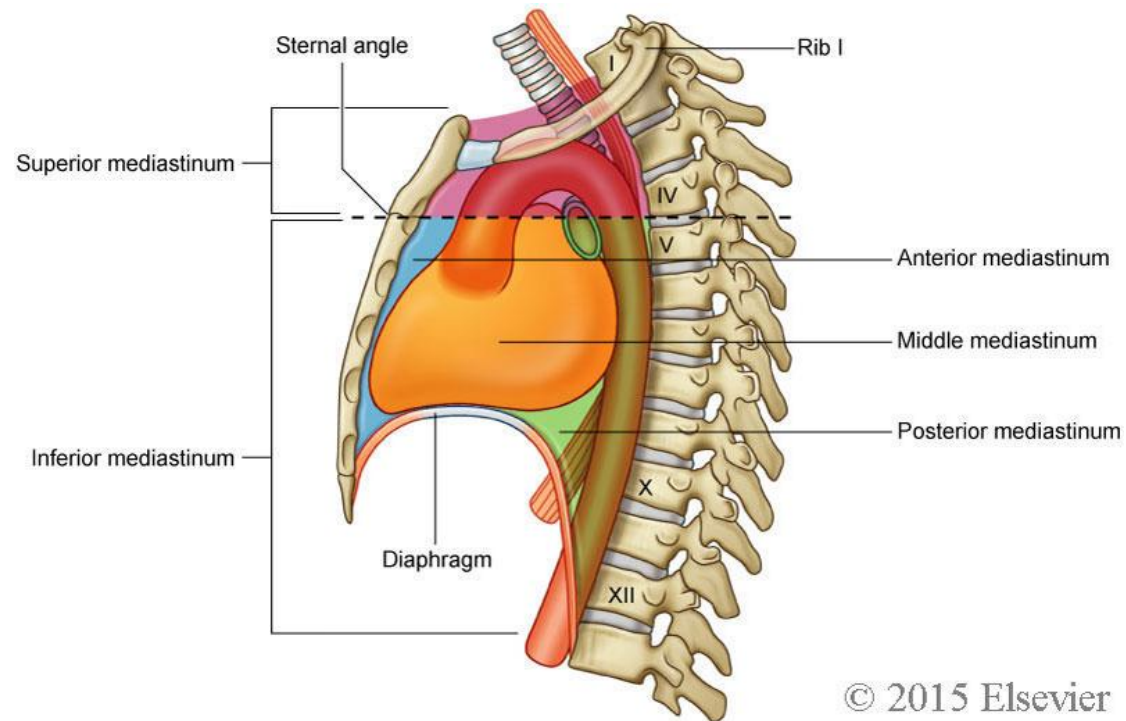
Inferior Mediastinum

Boundaries:

- Superior: transthoracic plane
- Inferior: diaphragm
- Anterior: body of the sternum
- Posterior: T5-T12

Divisions:

- Anterior
- Middle
- Posterior



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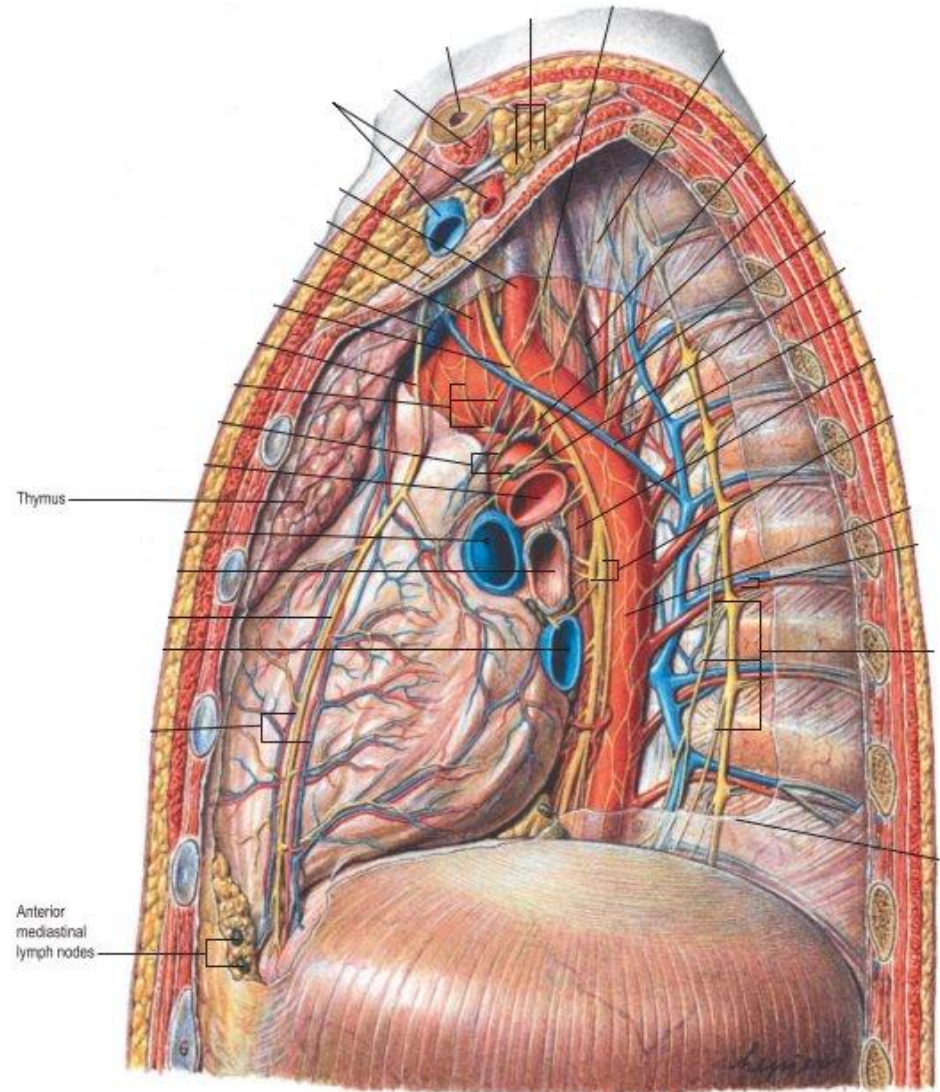
Anterior Mediastinum

Boundaries:

- Anterior: sternum
- Posterior: pericardium
- Superior: Transthoracic plane
- Inferior: Diaphragm

Contents:

- Thymus (or what remains)
- Lymph nodes
- Connective tissue



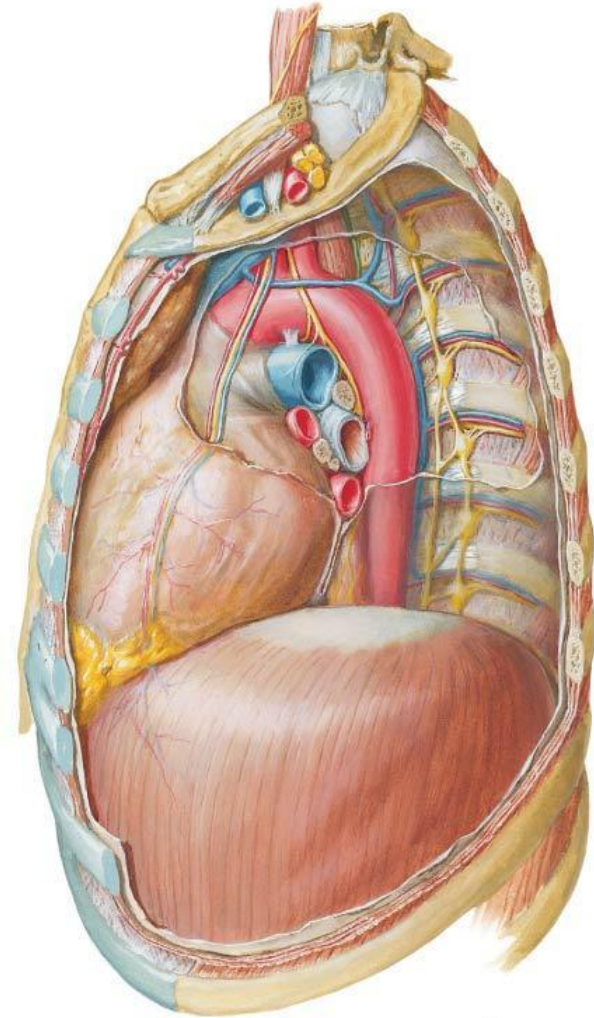
Middle Mediastinum

Boundaries:

- Bounded by the fibrous layer and parietal layer of serous pericardium on all sides.
- Pericardium is tightly adhered to the diaphragm inferiorly

Contents:

- Heart
- Ascending aorta
- Intrapericardial SVC
- Pulmonary trunk
- Main bronchi



Pericardium

- Functions: Limits acute cardiac dilation, assists atrial filling, mechanical and anti-friction protection
- Comprises two parts: **fibrous** and **serous** pericardia

Fibrous Pericardium

Connective tissue that attaches:

- Superiorly = **adventitia** of the great vessels
- Inferiorly = diaphragmatic tendon (**pericardiophrenic ligaments**)
- Anteriorly = sternum (**sternopericardial ligaments**)

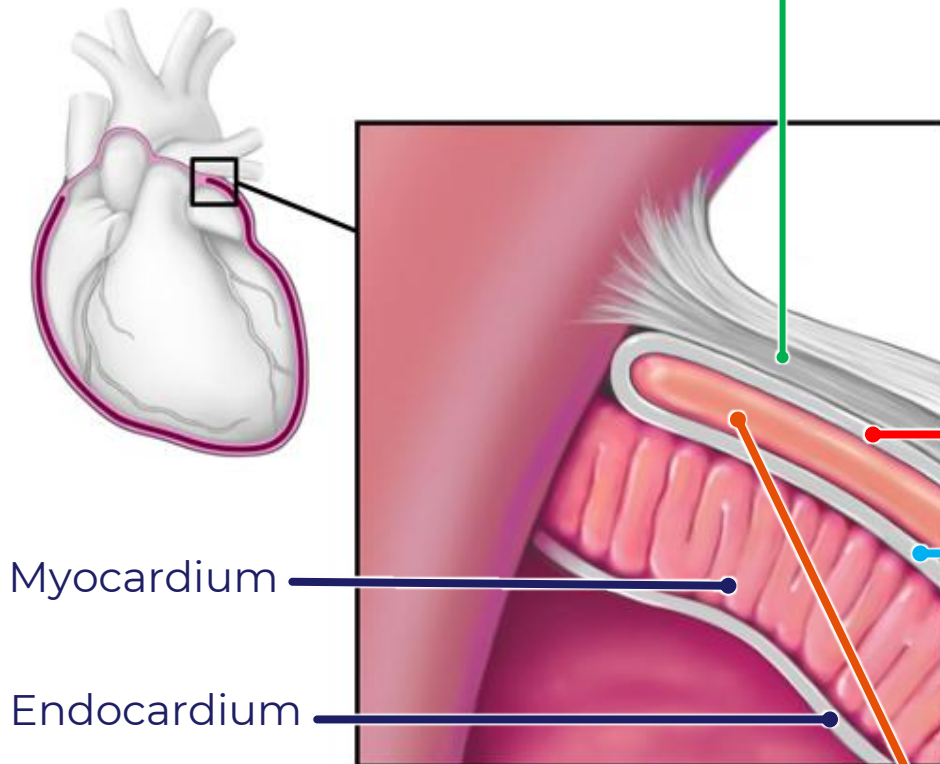
Serous Pericardium

Parietal – Outer (lines internal surface of fibrous)

Visceral – Inner (lines the heart, i.e. **epicardium**)

- Main coronary vessels and branches are embedded in the fat within epicardium

Separated by the **pericardial cavity** that contains a thin fluid film, 20 – 60 ml of serous plasma ultrafiltrate



Myocardium

Endocardium

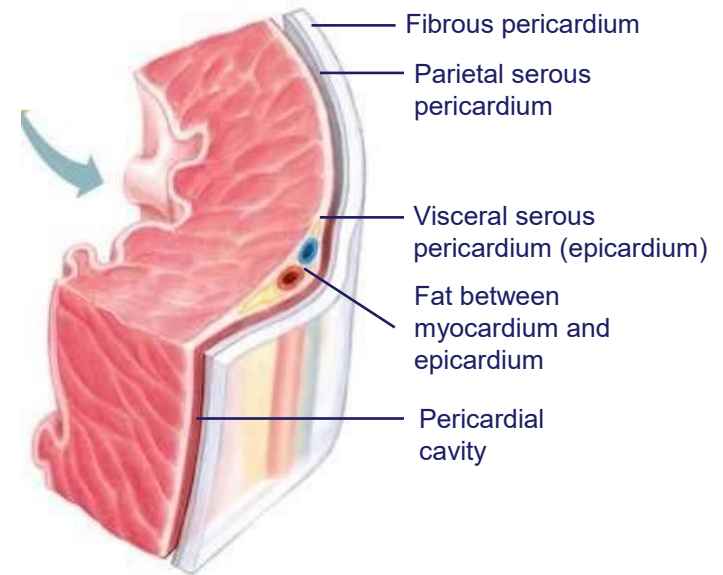
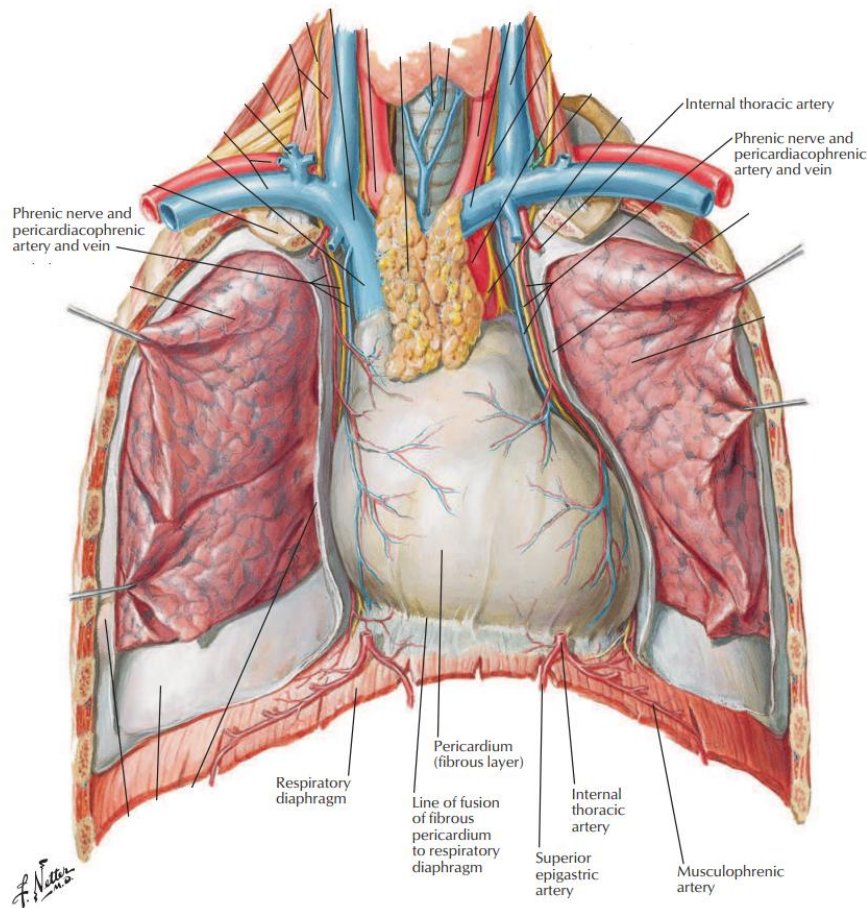
Pericardium - Blood Supply

• Venous drainage

- Azygos
- Internal thoracic
- Superior phrenic
- Cardiac (visceral serous)

• Arterial supply

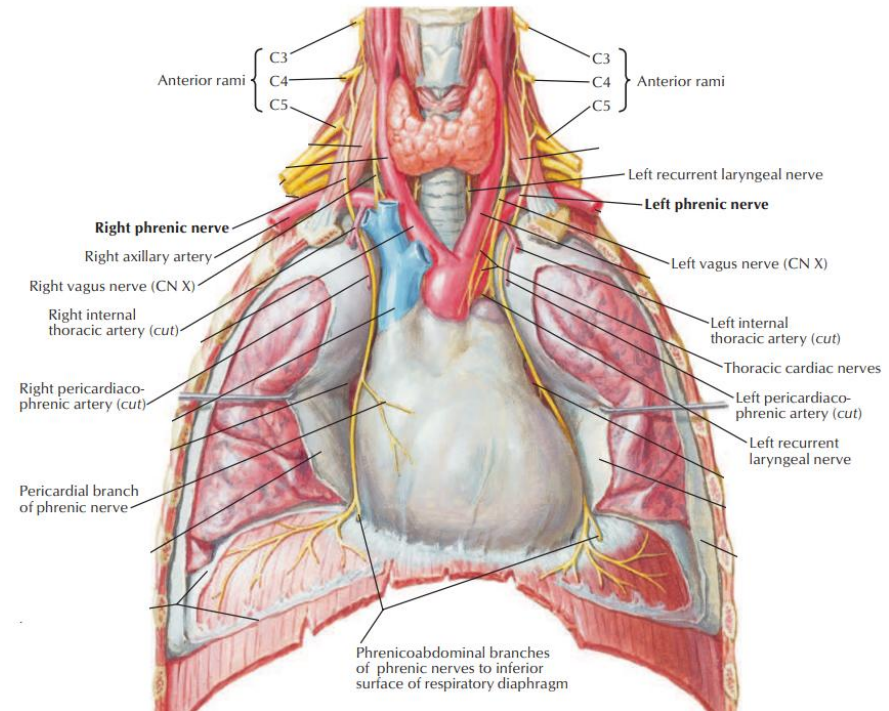
- **Pericardiophrenic (from internal thoracic)**
- Musculophrenic (from internal thoracic)
- Superior phrenic
- Intercostal
- Mediastinal (from thoracic aorta)
- Coronary (visceral serous)



Pericardium - Innervation

Parietal pericardium

- **Phrenic nerve**
- Somatic (pericardial) pain
 - Sharp and severe character
 - Substernal and radiates to shoulder
 - Exacerbated when lying supine
 - Relieved when leaning forwards



Visceral pericardium

- Sympathetic nerves (**aortic and cardiac plexuses**)
- Parasympathetic (**vagus nerve**)
- Visceral pain

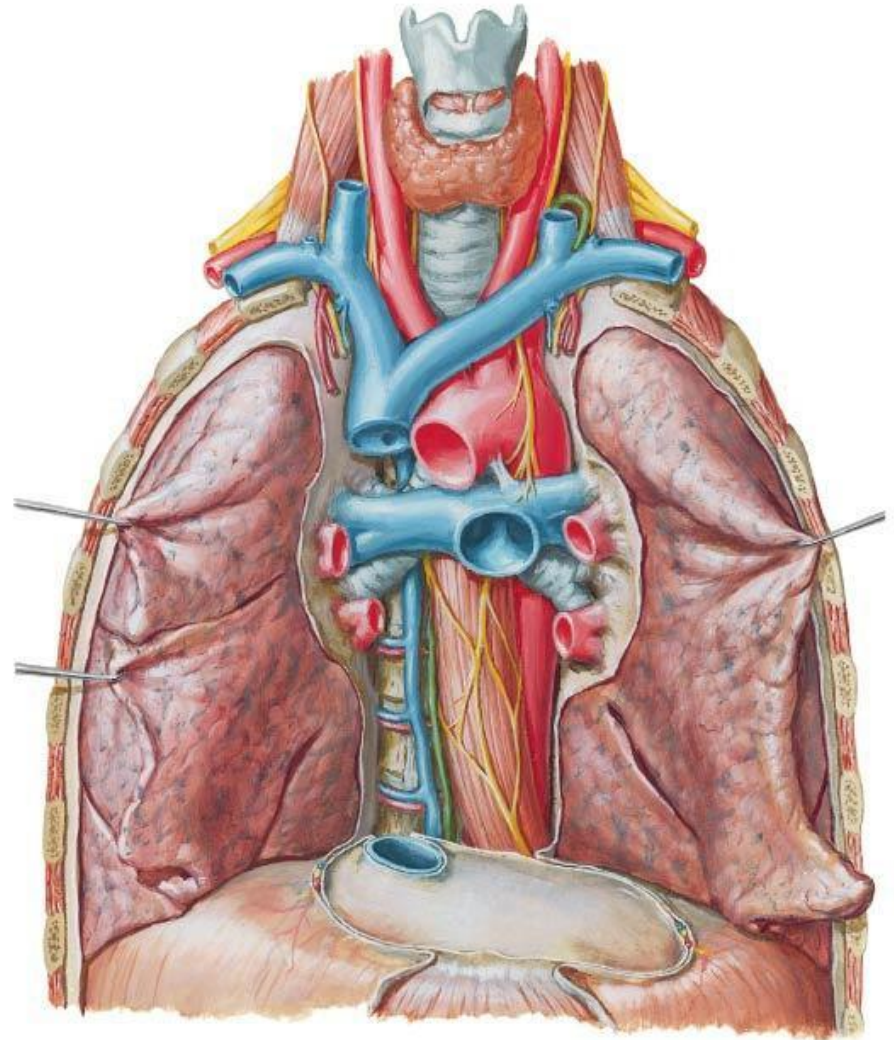
Posterior Mediastinum

Boundaries:

- Anterior: Pericardium
- Posterior: Vertebrae T5-T12

Contents:

- Esophagus
- Esophageal plexus***
- Descending thoracic aorta
- Azygos venous system
- Thoracic duct
- Sympathetic chain



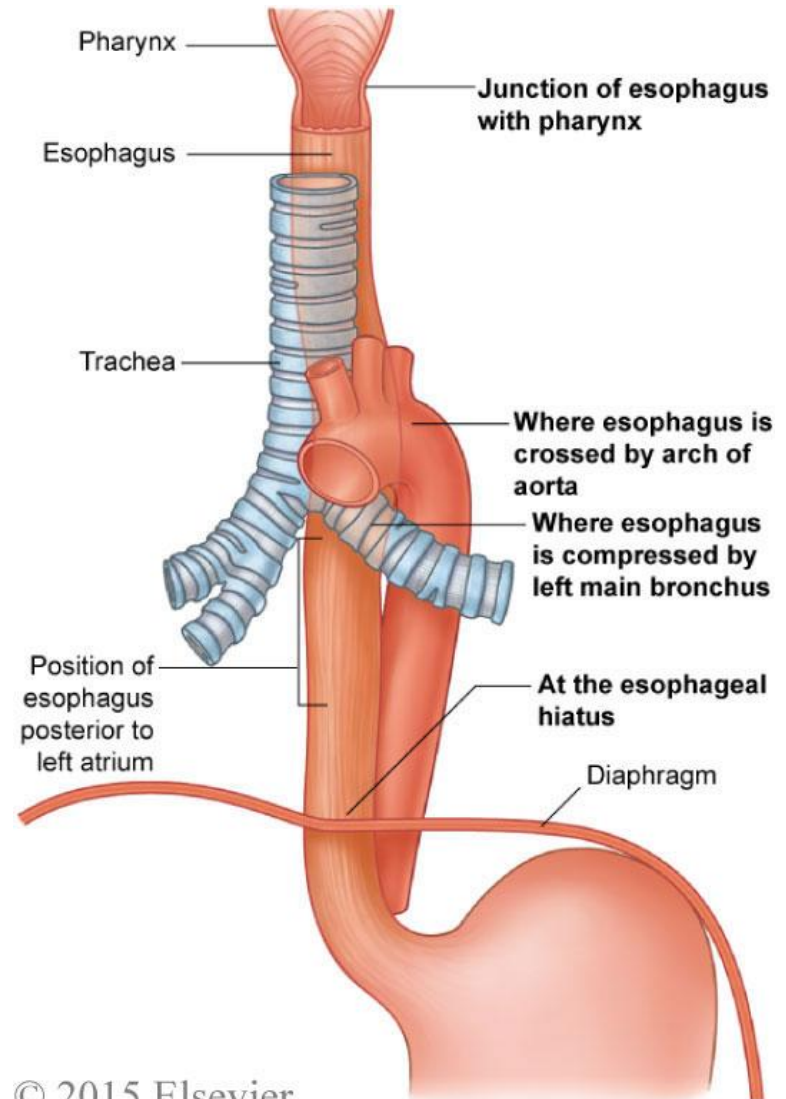
Esophagus

Location:

- Continuation of the pharynx from C6 vertebral level
- Enters the thorax through the superior thoracic aperture
- Passes through the diaphragm at **T 10 level**
- Ends at entrance of the stomach at the cardiac sphincter

Constrictions:

- **Pharyngoesophageal constriction** at the junction with the pharynx (C6)
- Crossed by **arch of aorta** (T4/5)
- Crossed by **left main bronchus** (T4/5)
- **Diaphragmatic constriction** as it passes through the esophageal hiatus (T 10)



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Esophagus – Thoracic and Abdominal

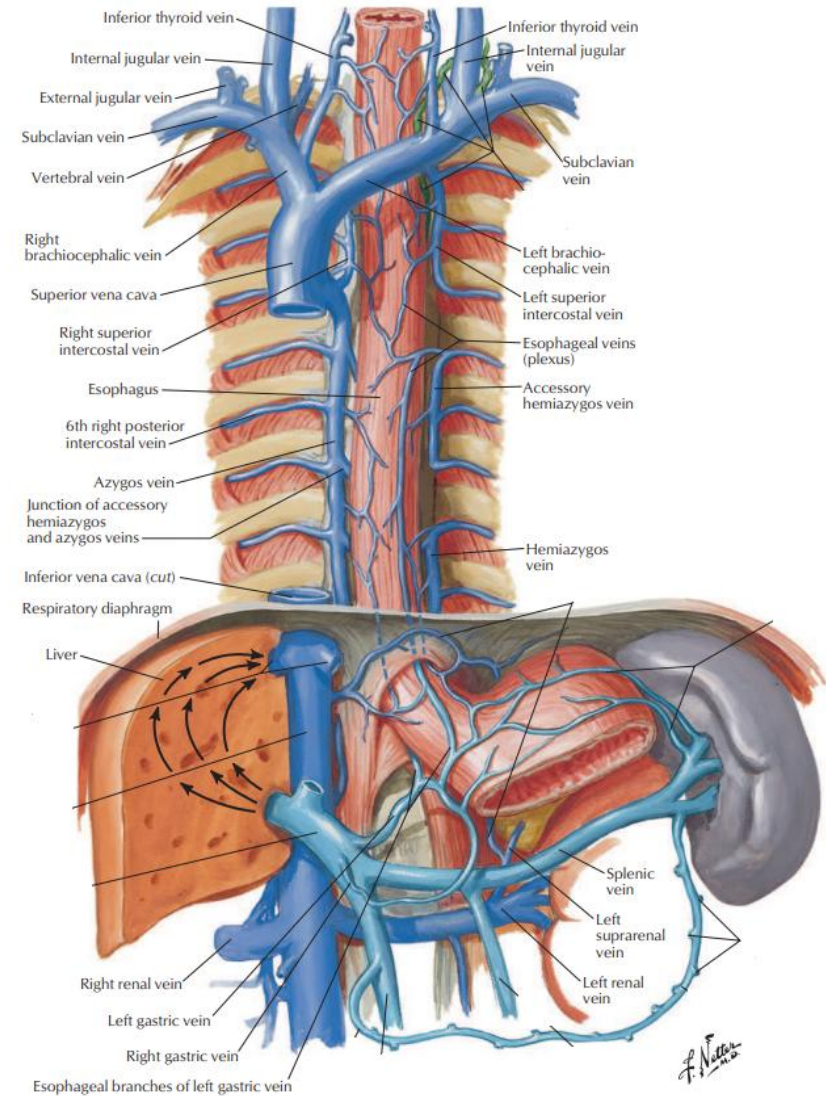
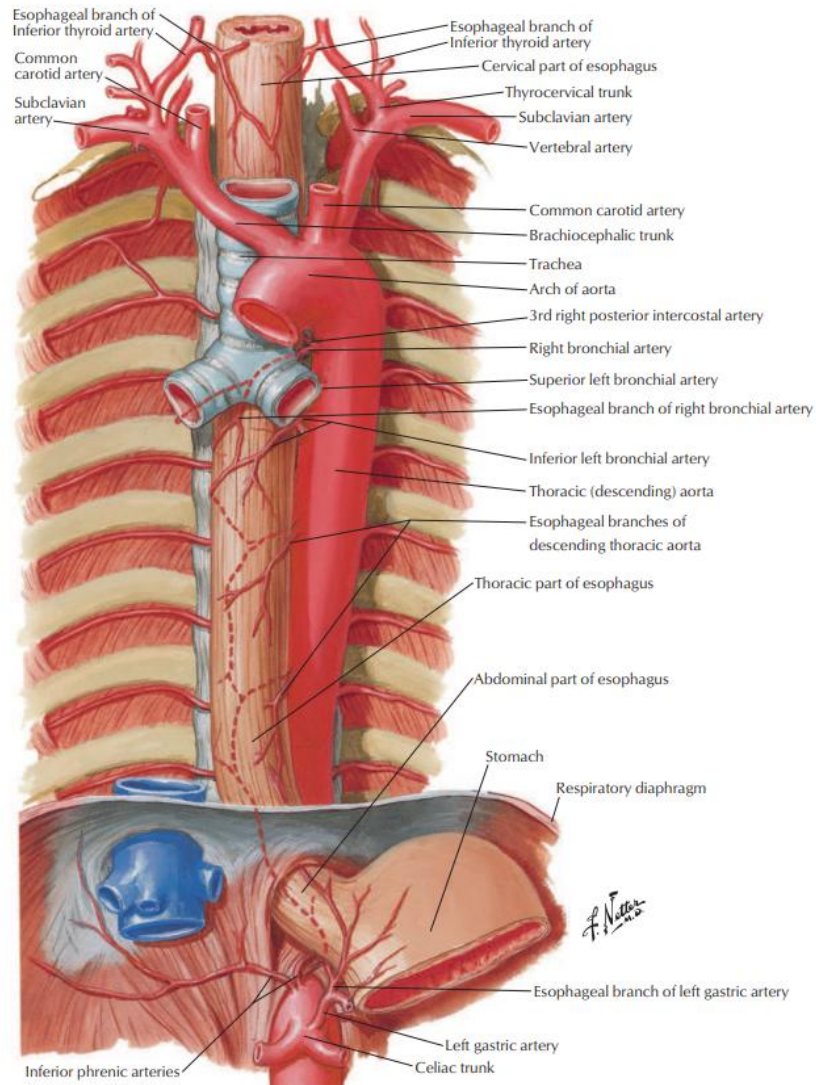
Thoracic Portion:

- Arterial supply: thoracic aorta & bronchial arteries
- Venous drainage: azygos, hemiazygos veins
- Lymphatic drainage: thoracic duct, posterior mediastinal nodes
- Innervation: esophageal plexus, sympathetic trunks

Abdominal Portion:

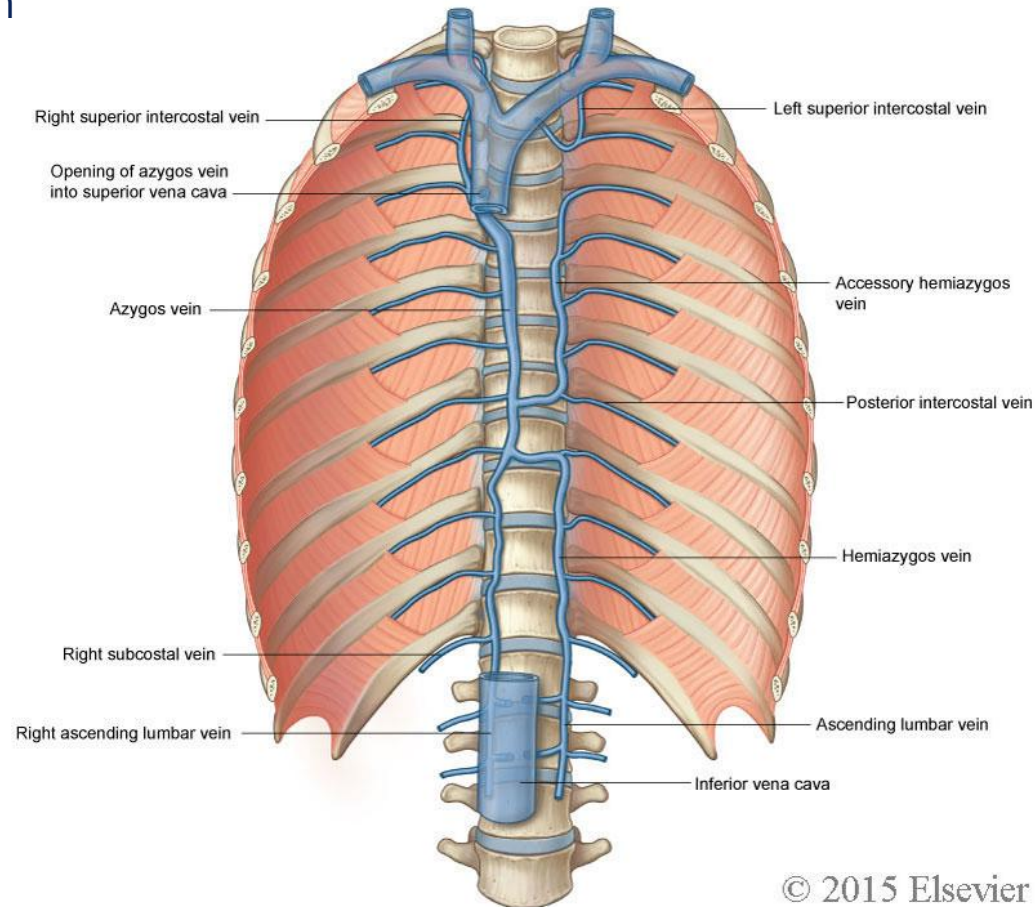
- Arterial supply: left gastric & left inferior phrenic arteries
- Venous drainage: left gastric vein
- Lymphatic drainage: left gastric nodes
- Innervation: vagal trunks, greater splanchnic nerves

Thoracic & Abdominal Esophagus



Azygos System of Veins

- Series of paired longitudinal vessels that drain blood from the body wall into the SVC
 - **Azygos**
 - **Hemiazygos**
 - **Accessory hemiazygos**
- Form collateral drainage pathway between IVC and SVC.

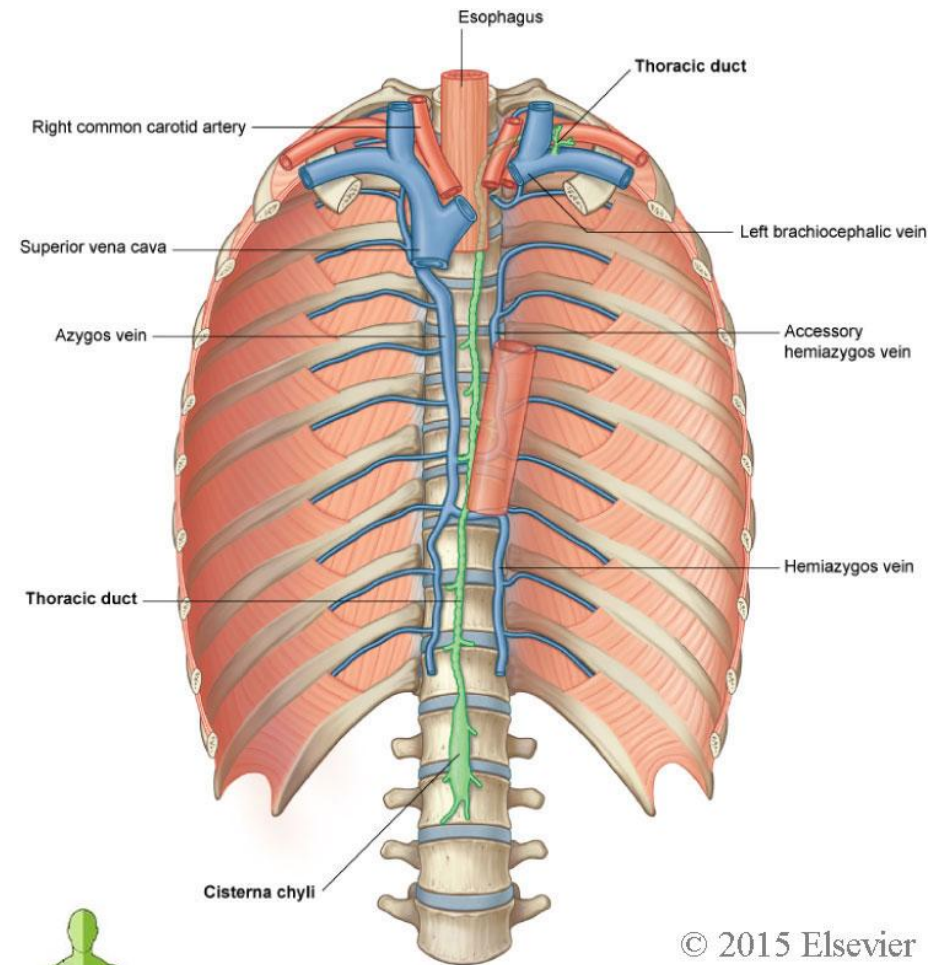


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Thoracic Duct

- Largest and most prominent lymphatic vessel in the body.
- Starts in the abdomen as confluence (cisterna chyli)
 - Passes through the diaphragm's aortic hiatus.
 - Crosses midline at T5 to the left side of the thorax .
 - Drains into the junction of left subclavian and left internal jugular veins.
- Carries lymph from the entire body except the right head and neck, right upper limb, and right thorax.



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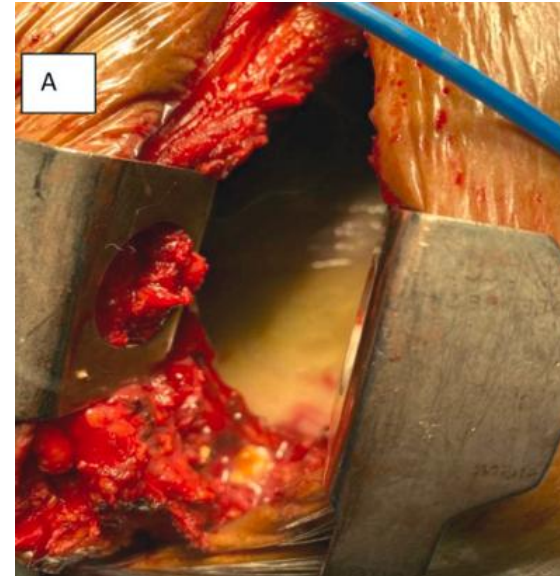
Injuries to the Thoracic Duct

Can be caused by:

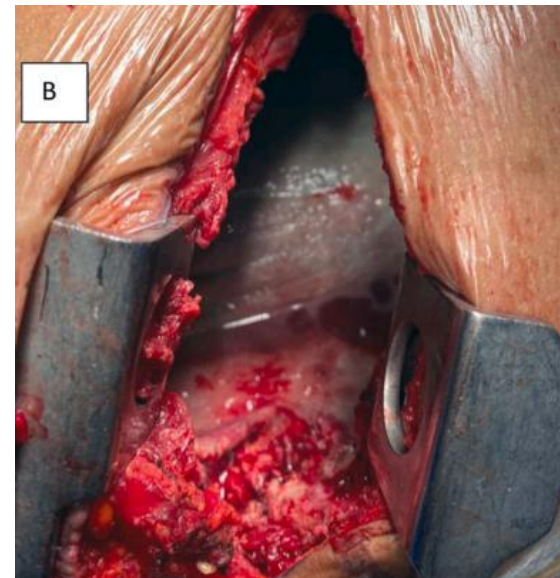
- Pathology affecting lungs, great vessels, esophagus, spine, lymph nodes
- Penetrating trauma, medical interventions
 - Catheterizations (e.g., internal jugular vein)
 - Surgeries (e.g., esophagostomy)

Presents with:

- Presence of chyle (fluid mixture of lipids, proteins, lymphocytes)
 - Chylothorax
 - Chylomediastinum
 - Chylopericardium



Right thoracotomy showing chyle



Thoracic duct ligated and clipped

DOI: 10.1016/j.amsu.2022.103623

Sympathetic Trunk & Splanchnic Nerves

- Sympathetic trunks are located laterally to the vertebral column
 - Contain interconnected sympathetic ganglia

Form 3 distinct splanchnic nerves:

Greater

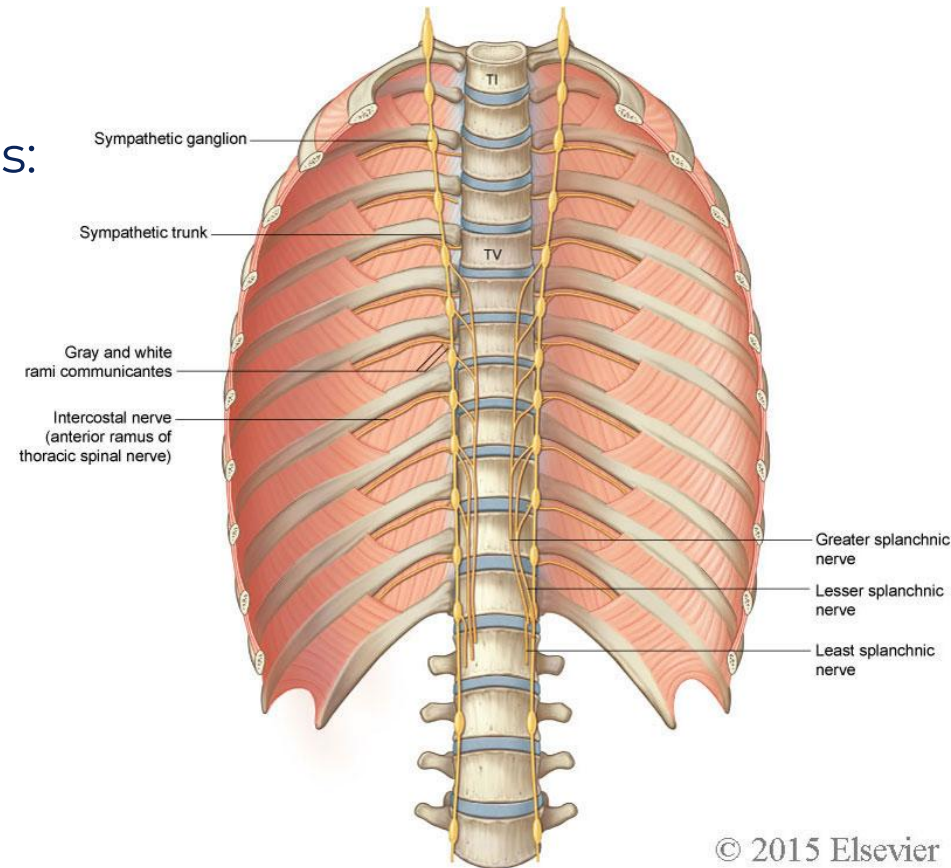
- T5-9 with some variation
- Joins the celiac ganglion

Lesser

- T10-11
- Joins aortico-renal ganglion

Least

- T12 but not always present
- Joins the renal plexus

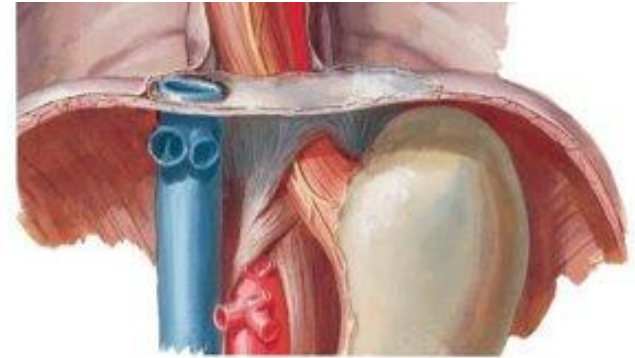


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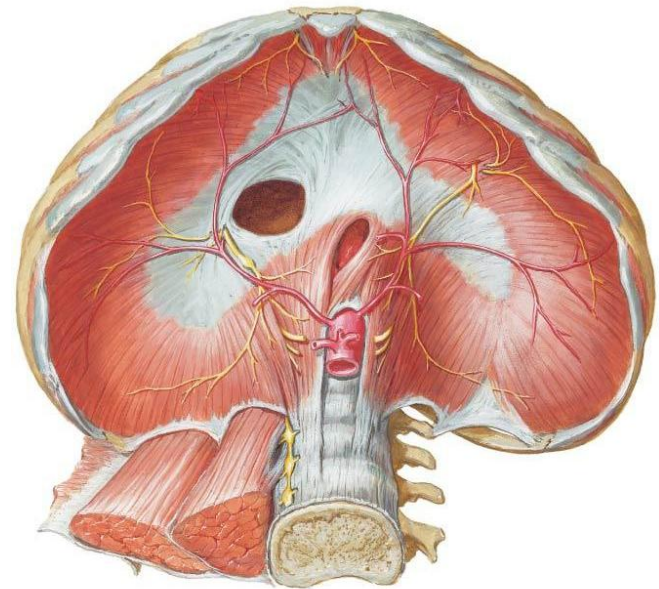
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Diaphragm

- A thin musculotendinous structure that fills the inferior thoracic aperture
- Separates the thoracic cavity from the abdominal cavity.
- Has three openings at the following levels:
 - T8 = Inferior vena cava (right)
 - T10 = Esophagus (left)
 - T12 = Aorta (midline)
- Peripheral attachments:
 - Xiphoid process of the sternum
 - Costal margin of the thoracic wall
 - Ends of ribs 11 and 12
 - Ligaments that span across the structures of the posterior abdominal wall
 - Vertebrae of the lumbar region



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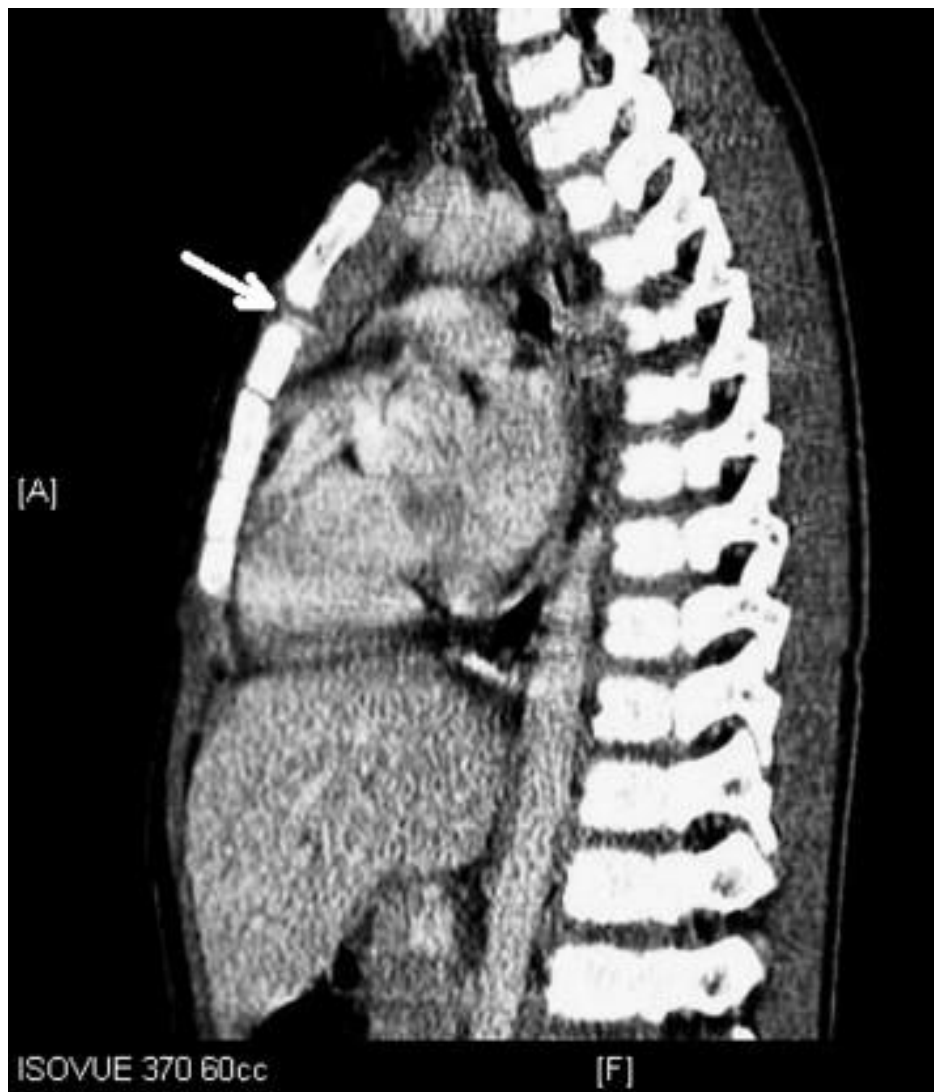
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Case Report - Application

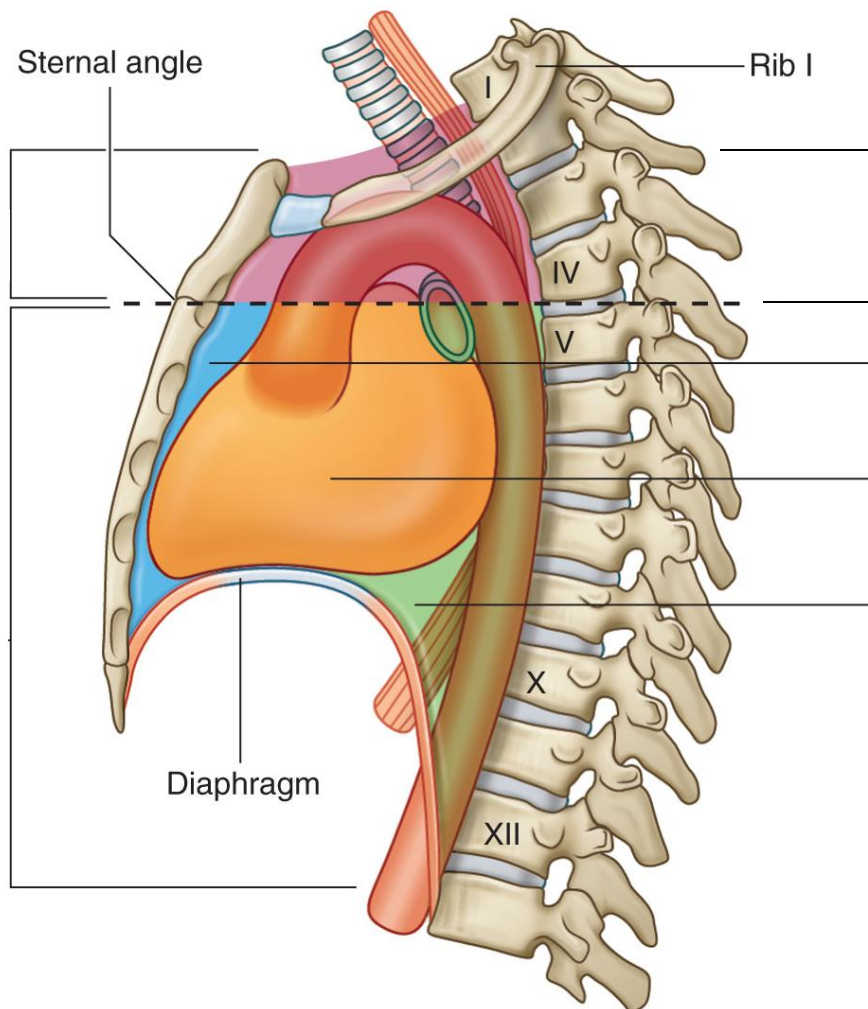
“barrel cactus”
(*Echinopsis pasacana*).



cactus spine entering the
anterior mediastinum



Summary



Superior mediastinum Arch of the aorta + its 3 branches, brachiocephalic veins, SVC, phrenic nerves, vagus nerves, left recurrent laryngeal nerve, thoracic duct, Thymus, trachea, esophagus, other small nerves, blood vessels, lymphatics.

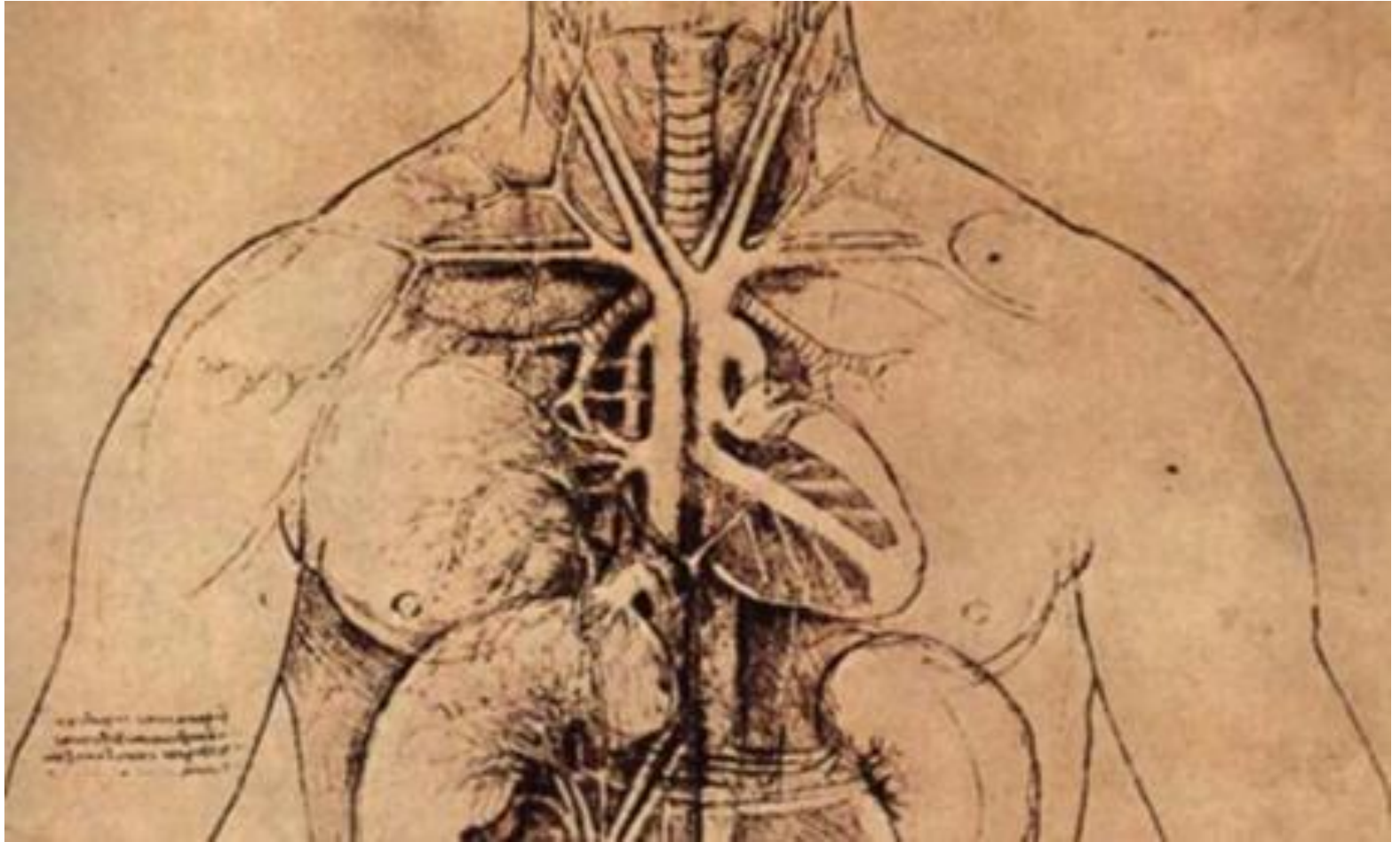
Anterior mediastinum Thymus, fat, connective tissue, lymph nodes, mediastinal branches of the internal thoracic vessels, sternopericardial ligaments.

Middle mediastinum Pericardium, heart, origins of the great vessels, main bronchi.

Posterior mediastinum Thoracic aorta + its branches, Azygos system of veins, esophagus + its nerve plexus, sympathetic trunks, thoracic splanchnic nerves, thoracic duct + lymph nodes.

Arteries, Veins, nerves, lymphatics, other structures

fin



from "Drawing of a Woman's Torso" by Leonardo da Vinci c.1507